

electronics, Network Security, and Cloud Computing, Optical Communication, R F Circuit Design.

11. Department of Environmental Science and Engineering

The Department of Environmental Science and Engineering have been offering M. Tech. in Environmental Engineering Degree and B. Tech in Environmental Engineering. The department conducts cutting-edge research, in developing the vital areas that address societal needs for environmentally sustainable life style and offer doctoral degree. In addition to this, the department also provides opportunity to working engineers for upgrading their qualification under Continuing Education Programme on part time basis for PG level. The departmental laboratories (Water Pollution, Air & Noise Pollution, Microbiology, GIS & Remote Sensing and Solid Waste) for teaching and research are among the best in the nation, providing opportunities for hands- on experience for all students.

Research Areas: Water Pollution, Waste Water Treatment, Environment Modelling, Phytoremediation, Water Management, Air Pollution, Geo- environmental Engineering, Solid Waste Management and Noise Pollution.

12. Department of Humanities

The Department of Humanities offers courses in Communication Skill, English, Economics and Accountancy for engineering and management students in an effort to train them for the global economic environment of the 21st Century. Besides giving them in-depth understanding of the labour market in which they are expected to work as well as of emerging employment trends among engineers, students are sensitized towards the specific technological needs of urban slums and rural areas and socio economic impact of engineering projects on the masses. The Department has competent faculty members with a high degree of excellence who keeps pace with the current developments in their fields of specialization for the fulfilment of its teaching and research goals. More than 20 research students are registered in the Department for Ph.D. programme.

Research Areas:

- a) Economics: Women Education and Inclusive Growth, Banking and finance, International Trade and other areas of

economics.

- b) English: Contemporary Fiction, Cultural Studies, Diaspora Studies, Post-modern literature.

13. Department of Information Technology

The Department of Information Technology (IT) endeavors to provide the thrill of a corporate R&D environment with a planned focus on industrially relevant subjects, projects and technology incubation. The department offers an undergraduate (B. Tech.) course in Information Technology with an intake of 180 students every year. To meet the growing demands of present day technologies, the department has started M.Tech. degree in Information Systems in 2009-10 with an intake of 25. The curriculum of the department is designed in a way so as to provide the students with fundamental concepts and learning tools related to outcome based studies. The designed courses mainly emphasize on all basic core subjects such as data structure, operating systems, computer architecture and design, software development, networking, multimedia and graphics, analog and digital communications and computer communications, compiler design, theory of computation, etc. The department is also imparted a specialized subject knowledge on analysis and design of Information system, Information security systems, mobile computing, Internet of Things, cloud computing & security, soft computing, artificial intelligence, digital signal processing, computer vision and expert systems and web engineering. The department has developed the various state-of-the-art laboratories in the fields of Information Technology such as Computer Networks Lab, Web Engineering Lab, Programming Lab, Information and Security Lab, Biometric Research Lab, and many others.

Research Areas: Pattern Recognition, Computer Vision, Soft computing, Biometric security system, Neural Networks/ Deep learning, Fuzzy-Neural Networks, Natural Language Processing, Optimizations Techniques, Computer Vision, Big data Analytics Web Mining Internet Technologies, Data Mining Social Networks, Social Media Mining, Social Computing, Human behaviour, Multimedia Systems Human Computer Interaction (HCI), Image

processing, Human Action and Activity Recognition, Sentiment Analysis, Spam Analysis, Fake News Analysis, Rumour Detection, Evolutionary Computing, Wireless Ad-hoc & Sensor Networks, Internet of Things (IoT), Software Defined Networking (SDN), Network Security, Information Security, Mobile Security, Internet of Robotics Things (IoRT), Cyber Physical System Security, Flying Ad-hoc Network (FANET) Security, Distributed Computing, Pattern Mining and Digital Forensics, Blockchain Technology, Recommendation Systems, Affective Computing, Autonomous Vehicles, etc.

14. Department of Mechanical Engineering

The Department of Mechanical Engineering also offers undergraduate and Postgraduate courses with specialization in

- a) Thermal Engineering
- b) Production Engineering.
- c) Industrial Engineering and Management
- d) Computer Aided Analysis and Design
- e) Energy Systems and Management

Ph. D Programmes in all fields of Mechanical Engineering are also offered. The department possesses modern laboratories equipped with latest experimental set-ups and research facilities for instrumentation, experimental stress analysis, strength of materials, fluid mechanics, tic, engines, automotive engineering, robotics, heat transfer, solar energy, flexible manufacturing system, computational fluid dynamics supported by Software like view-flex, CAD-CAM and I.e. engine design. Cad lab has Softwares like NX-LAD, NXCAM, AUTOCAD Inventor, Catia, Techomatix, Abus, Iadino, NX-Narran, Hyper mesh, hyper works, MDADAMS, Dynaform etc. The department has many research projects which are sponsored by different government organizations.

Research Areas: Turbo Machinery, Fluid Mechanics, Power Plant Engineering, Refrigeration and Air conditioning, Computational Fluid Dynamics, Solar Energy, Bio Fuels, Power Plant, Industrial Engineering & Supply Chain Management, Robotics, CAD/CAM, Welding, Production Engineering, System Dynamics, Structural Vibration, Modeling & Simulation, Automation, Advanced Manufacturing Process, Human Factor Engineering, Quality Engineering.

15. Department of Software Engineering

The Department of Software Engineering is dedicated to produce high quality graduates and skilled software engineers/professionals who can develop high quality and cost-effective software systems.

The Discipline of Software Engineering was introduced in the year 2009. The Department is currently running a B.Tech program in Software Engineering with an intake of 180, two M.Tech programs in Software Engineering and Data Science and offers Ph.D. in the Discipline of Computer Science and Software Engineering. All the software engineering programs are well designed keeping in view the industry demands. The programs are designed to build the analytical and practical capabilities of students in the design and development of the software and lays emphasis on following well defined and systematic approach for meeting the growing demands and requirements of the software industry. The department has state of art labs consistent with industrial standards which provide a hands-on experience to the students.

Research Areas: Empirical software engineering, machine learning, software quality and testing, search based software engineering, web engineering, opinion mining, social web, predictive modeling, machine learning and Deep learning for mobile healthcare, telemedicine, internet of things, cryptography.

16. USME

USME (University School of Management and Entrepreneurship) The Vision of the USME is to develop and nurture the spirit of management leadership and entrepreneurship for the good of society. The mission focuses on a portfolio of Programmes around entrepreneurship and cutting edge areas in management, and offers courses at the undergraduate and postgraduate levels: Executive MBA (Data Science and Analytics MA (Economics), BA (Hons.) Economics and BBA (Hons.) four year degree and BBA (IEV).

The focus of the department is to create a practice school in the area of management which is based on research and leads back to Programmes with a strong focus on entrepreneurship, employability, skill